

Drug Shortage—Associated Increase in Catheter-Related Blood Stream Infection in Children

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KEY WORDS

ethanol lock therapy, short bowel syndrome, catheter-related blood stream infection

ABBREVIATIONS

CRBSI—catheter-related blood stream infection

CVAD—central venous catheter device

FDA—Food and Drug Administration

IF—intestinal failure

PN—parenteral nutrition

SBS—short bowel syndrome

Dr Ralls is the primary author and contributed to conception and design of the study, data collection, and analysis and interpretation of data; Drs Blackwood and Arnold contributed to conception and design of the study, analysis and interpretation of data, and manuscript revision; Dr Partipilo contributed to data collection and to analysis and interpretation of data; Mr Dimond contributed to analysis and interpretation of data and drafting of the manuscript; and Dr Teitelbaum was the primary investigator; contributed to conception and design of the study, analysis and interpretation of data, and manuscript revision; and gave final approval for publication.

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abstract

BACKGROUND: Ethanol lock therapy (ELT) has been shown to reduce the incidence of catheter-related blood stream infections (CRBSI) in intestinal failure (IF) patients. Dosing and frequency remains undefined. Scrutiny of pharmaceutical facilities by the Food and Drug Administration led to the voluntary shutdown of the sole supplier of ethanol, resulting in a nationwide shortage. To conserve supply, we reduced ELT frequency from a daily regimen. We examined the impact that reduction in ELT frequency had on CRBSI in pediatric IF patients.

METHODS: We retrospectively reviewed our parenteral nutrition-dependent IF children. Primary outcome measure was CRBSI per 1000 catheter days after ELT frequency reduction. Data were compared (paired *t* test) to the same group over 1 year before ethanol shortage and to historical controls.

RESULTS: During the shortage 13 outpatients received ELT. Eight met study criteria. Mean $\pm$ SD age was 9.1 ± 7.8 years. Mean CRBSI rate per 1000 catheter days was 0.7 ± 1.3 before ELT shortage. This increased to 6.2 ± 2.5 after frequency reduction ($P < .001$). This CRBSI rate was similar to historical IF children not on ELT (8.0 ± 5.4). Seven children developed CRBSI after frequency reduction, 6 requiring hospitalization, 2 to the ICU. Mean length of stay (15.5 days) averaged \$104,783 ($\pm 111,034$) in hospital charges. Organisms included Gram-negatives (6), methicillin-resistant *Staphylococcus aureus* (1), and *Candida* spp (1).

CONCLUSIONS: ELT frequency reduction resulted in complete failure in CRBSI prophylaxis. The nationwide shortage of this drug has been costly both financially and in patient morbidity. *Pediatrics* 2012;130:e1369–e1373

Intestinal failure (IF) in the pediatric population is most commonly a result of motility disorders, congenital diseases of enterocyte development, and short bowel syndrome (SBS)¹; its incidence is estimated to be between 0.2 to 4.9/million.² Most pediatric IF/SBS cases are due to necrotizing enterocolitis, gastroschisis, omphalocele, intestinal atresia, and midgut volvulus.³ Many of these patients require resection of much of their small bowel, leaving insufficient surface area to absorb nutrients required for normal growth and development.⁴ Often, they require central venous access devices (CVAD) for long-term parenteral nutrition (PN).

One of the major morbidities associated with PN administration is catheter-related blood stream infection (CRBSI).⁵ The incidence of CRBSI is widely variable in the literature but reported to be between 16 000 and 500 000 annually in the United States alone.⁶ These infections often result in significant morbidity and mortality for the individual with the average admission costing between 3700 and 56 000 US dollars.⁷

Ethanol lock therapy (ELT) has been used to decrease CRBSI incidence.^{8,9} In some studies, the rate of CRBSI was reduced by as much as 90%.^{10–13} The Centers for Disease Control and Prevention Guidelines for the Prevention of Intravascular Catheter-Related Infections now recommends the use of prophylactic antimicrobial lock solution in patients with long-term catheters.¹⁴ ELT has been applied to the high-risk population of pediatric patients with IF to prevent CRBSI with excellent success.^{15,16}

Recent scrutiny of pharmaceutical production facilities by the US Food and Drug Administration (FDA) led to the voluntary shutdown of the sole supplier of 98% dehydrated ethanol, resulting in a nationwide shortage of ethanol for lock applications.¹⁷ The temporary

suspension of manufacture and distribution of many products, including dehydrated alcohol, began in April 2011,¹⁸ and allocation efforts continued through September 2011. A conservation plan for dehydrated alcohol dosage frequency was developed for our patients in an effort to maintain as many patients on ELT as possible. Daily ethanol lock dosing was resumed once sufficient quantity of product was received. We hypothesized that the reduced ELT frequency would lead to a significant increase in the incidence of CRBSI, leading to increased patient morbidity and associated health care costs.

PATIENTS AND PRESENTATION

At the time of the ethanol shortage, 8 patients underwent a reduction in the frequency of their ethanol instillation from daily to less than daily in an attempt to ration our remaining supply. These children were then followed throughout the shortage and the rate of CRBSI per 1000 catheter days during decreased frequency was compared with the same individuals' rates of infection from the year before the shortage using a 1-tailed paired *t* test. As an additional approach to analysis, rates of CRBSI were also compared with historical controls of patients with IF both with and without daily ethanol use over the 7 years. CRBSI was defined, using the Centers for Disease Control and Prevention 2002 guidelines,¹⁹ as a positive blood culture obtained from a peripheral vein and/or central line in a patient demonstrating systemic symptoms. The study was approved by our hospital's institutional review board (protocol HUM00053172). The mean ($\pm$ SD) age of the patients at the time of change in ELT frequency was 9.1 ± 7.8 years (range 22 months–18 years). The diagnoses leading to IF included 3 patients with necrotizing enterocolitis and 2 patients with

gastroschisis. Other causes included 1 patient with Hirschsprung disease with a complicated course, 1 child with congenital short bowel, and 1 patient with gastrointestinal dysmotility of unclear etiology.

The mean duration of ELT therapy at the time of shortage was 1000.8 ± 563.8 days (range 309–1732 days). Six patients were decreased from daily ELT to twice weekly ELT, 1 patient to weekly, and 1 patient received ELT less than once weekly in that there was a brief time period during which no ELT was administered. These regimens were based on our unallocated supply as well as the previously distributed supply of each individual patient.

All patients had tunneled silastic catheters. Six patients had single lumen catheters; 2 children had a double lumen catheter. Those with double lumen catheters received ELT in each lumen on alternating days before the shortage and ELT once weekly in each lumen thereafter. The duration that the patients' current catheters had been in place at the time of reduction was variable with a mean of 1049 ± 1103 days (range 107–3392).

The rate of CRBSI in these 8 patients in the year before the ethanol shortage matched previously published data.¹⁶ After the shortage, 7 patients developed a CRBSI during the study period (Table 1). The overall rate of CRBSI per 1000 catheter days after frequency reduction increased to 6.16 versus 0.68 ($P < .001$). This rate of CRBSI after ELT reduction was slightly lower than historical IF patients not receiving ELT (8.0 ± 5.4) but was not statistically significant ($P = .27$).

Six of the 7 patients diagnosed with CRBSI required hospitalization and intravenous antibiotics. Mean length of stay for each admission was 15.5 ± 9.2 days (range 6–28 days). Two patients required ICU admission for septic shock. One child responded well to

TABLE 1 Mean Rate of CRBSI per 1000 Catheter Days $\pm$ SD

	Pre-ELT (historical)	Daily ELT (historical)	Daily ELT	Less than daily ELT ^a	P value
Daily ELT _{Historical} vs daily ELT _{Current}	8.0 $\pm$ 5.4	1.3 $\pm$ 3.0 1.3 $\pm$ 3.0	0.68 $\pm$ 1.27 0.68 $\pm$ 1.27	6.16 $\pm$ 2.5	.55
Daily ELT _{Historical} vs less than daily ELT		1.3 $\pm$ 3.0		6.16 $\pm$ 2.5	<.001 ^b
Daily ELT _{Current} vs less than daily ELT			0.68 $\pm$ 1.27	6.16 $\pm$ 2.5	<.001 ^b
Pre-ELT _{Historical} vs less than daily ELT	8.0 $\pm$ 5.4			6.16 $\pm$ 2.5	.27

^a Less than daily is defined as either twice weekly or weekly administration.

^b Statistically significant.

therapy and was in the ICU for 2 days, whereas the other required prolonged intubation and a 20-day ICU stay. His course was subsequently complicated by respiratory failure with hypoxia secondary to severe postintubation subglottic stenosis requiring readmission and emergent tracheostomy placement. In addition, 6 children were put under general anesthesia for CVAD removal and/or replacement.

Interestingly, during the shortage, there was a preponderance of Gram-negative organisms (7 of 9 isolates) compared with either the preshortage or historical controls (Table 2). However, all isolated organisms are known to colonize the GI tract routinely.

An average of \$104 784 ($\pm$ 111 034) in charges was accumulated per admission. However, these totals range greatly from \$16 000 per encounter to

>\$232 000 in 1 case. This value was widely variable based on the severity of the infection and the resultant comorbidities. A significant amount of incremental costs were also incurred. As an example, 1 patient experienced >\$3000 in ventilator therapy costs and >\$9300 in isolation expenses.

DISCUSSION

The use of daily ELT has been gaining national acceptance. We were reluctant to modify our ELT protocols given our favorable experience and the successes of other institutions mirroring our own. However, it would be prudent to continue daily ELT and rapidly deplete our critical supply. Although we were concerned that a reduction in frequency would cause an increase in CRBSI, it was startling that the decrease in frequency from daily dosing to weekly and

semiweekly administration led to the complete failure in ELT efficacy.

This report demonstrates the profound health implications due to the shut-down of a pharmaceutical company, particularly when it is the sole provider of a single agent. The impact on children in terms of morbidity and financial costs were far beyond our initial expectations when we planned for this shortage. Although a number of drug shortages have resulted in inconveniences and difficult work-arounds in the field of nutrition support, the adverse health implications of the loss of ethanol has had the greatest adverse effect on our children.

Children with IF on PN are at the highest risk of numerous complications, including CRBSI, that far surpass the rates in the adult population.^{20,21} Children with SBS have a higher incidence of CRBSI than children with CVADs for other diagnoses.²² These infections are more often caused by enteric organisms,²³ suggesting that bacterial translocation may be the cause of CRBSI in this subset. In our IF children receiving daily ELT, 40% of the blood culture isolates were *Staphylococcus aureus*, and 40% were *Staphylococcus epidermidis*. In contrast, once ELT was reduced to less than daily, enteric organisms became the most common pathogen

TABLE 2 Causative Organisms in CRBSI in Children With Short Bowel Syndrome

	Pre-ELT (historical)	Daily ELT (historical and current data)	Less than daily ELT
Gram-negative organisms	<i>Klebsiella pneumoniae</i> (7); <i>Citrobacter spp</i> (2); <i>Escherichia coli</i> (2); <i>Enterobacter cloacae</i> (1); <i>Klebsiella oxytoca</i> (1); <i>Morganella morganii</i> (1); <i>Acinetobacter baumannii</i> (1); <i>Stenotrophomonas maltophilia</i> (1)	<i>Pseudomonas aeruginosa</i> (1) ^a	<i>K. pneumoniae</i> (2); <i>E. coli</i> (1); GNR (unspecified) (1); <i>P. aeruginosa</i> (1)
Gram-positive organisms	<i>Staphylococcus epidermidis</i> (14); <i>Enterococcus faecalis</i> (9); <i>Vancomycin-resistant enterococci</i> (2); α -hemolytic <i>streptococci</i> (1); <i>Bacillus cereus</i> (1); <i>Enterococcus faecium</i> (1); <i>Kocuria kristinae</i> (1); <i>Staphylococcus aureus</i> (1); <i>Staphylococcus spp</i> (1); <i>Candida albicans</i> (4); <i>Candida parapsilosis</i> (2)	<i>S. aureus</i> (4); <i>S. epidermidis</i> (4)	<i>Enterococcus</i> (1) ^a ; <i>S. aureus</i> (1)
Other organisms		<i>Candida lusitanae</i> (1) ^a	<i>C. lusitanae</i> (1) ^a

GNR, Gram Negative Rod.

^a Species isolated from same individual within the same hospitalization.

isolated. The causative organisms in our study population resembled that of the historical pre-ELT group.

Children with SBS have a high rate of complications and have multiple potential reasons for illness and readmission.^{21–23} Chronic complications stemming from PN include CRBSI, venous thrombosis, metabolic complications, anemia, hepatic dysfunction, and demineralization of bone and renal stones.²⁴ Repeated infections have been associated with worsening of PN-associated liver disease.²⁵ The chronically ill patient may have a decreased ability to tolerate these complications. Any intervention that decreases this population's rate of complications and hospitalization is of critical importance.

The FDA has the difficult task of balancing under- and overregulation. Manufacturers must meet high standards for new products to ensure public safety. It is argued that overregulation is difficult to publicize, whereas consequences of approving a drug that is later proven to be harmful will have a detrimental impact on the FDA.²⁶

In 2010, there was a record number of shortages, and there was an increasing number of shortages in 2011.²⁷ The FDA cannot require companies to report the reason for or anticipated duration of the shortage. Consequently, the precise details that led to the voluntary shutdown of the sole supplier of 98% dehydrated ethanol and the subsequent shortage of medicinal grade ethanol were not transparent. However, it is inferred that the threat of

an impending FDA inspection was the catalyst to the closure of this company's production line, which led to a dramatic increase in the morbidity of our PN-dependent IF patients. Clearly, both industry and the FDA must find a safer way to allow an uninterrupted supply of such critical pharmaceuticals to the patients who need them.

The rationing that resulted from the shutdown led to a complete failure of the prophylactic properties of ELT that has been demonstrated in this study and others. The detrimental health impact on our patients and their families was great. It is hoped this report will help influence legislature, industry, and the FDA to work to provide a stronger safety net for our children who are reliant on critical and lifesaving medications.

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